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RM3100 Frequently Asked Questions

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Draft	Initial Version Q1 to Q13	3/16/2018
R01	Q14	4/10/2018
R02	Add more Qs for a total of 17	11/12/2019
R03	Add Q18	12/11/2019
R04	Add Q19	01/09/2020
R05	Add Q20	2/13/2020
R06	Add Q21-23	10/22/2020

1 WHAT IS RM3100?

RM3100 is PNI 3-axis magnetic sensor suite. It measures magnetic field. It consists of 4 components, MagI2C ASIC, two X-axis or Y--axis sensors, and one Z-axis sensor.

2 WHAT ARE RM3100 INTERFACE PORTS?

RM3100 supports I2C and SPI. RM3100 is I2C or SPI Slave to communicate to a host. Basic schematics for implementing the RM3100 Sensor Suite when using the SPI interface and I2C interface are provided in RM3100 user manual.

3 WHAT IS CYCLE COUNT AND HOW TO CHANGE IT?

The internal clock count of RM3100 MagI2C ASIC establishes the number of sensor oscillation cycles. The number of oscillation cycle is “Cycle Count”.

Cycle Count as default value is 200 or 0xC8 in hex. Each axis can have different cycle count, to change, set CCX, CCY and/or CCZ registers. They are 16-bit registers.

4 WHAT IS THE RELATIONSHIP BETWEEN CYCLE COUNT AND GAIN?

Cycle count and gain are linearly related. See the following table on “Cycle Count” and “Gain”.

Cycle Count/Gain/Sensitivity			RM3100 Measurement	
Cycle Count	Gain (LSB/uT)	Sensitivity(nT/LSB)	in count	microTesla (uT)
50	20	50.000	3000	150.000
100	38	26.316	3000	78.947
200	75	13.333	3000	40.000
300	113	8.850	3000	26.549
400	150	6.667	3000	20.000

5 WHAT IS THE RELATIONSHIP BETWEEN GAIN AND SENSITIVITY?

Sensitivity is the inverse of gain, Sensitivity = 1/Gain. See Q4 Table “Gain” and “Sensitivity”.

6 HOW TO READ OUT RM3100 MEASUREMENT ?

RM3100 measurement is a 24-bit signed integer in count.

```
//read out measurement from register 0x24 to the char mSamples[9] array
rm3100_i2c_read(Data_REG0X24, (char*)&mSamples, sizeof(mSamples)/sizeof(char));

SInt32 XYZ[3];
XYZ[0] = ((signed char)mSamples[0]) * 256 * 256;
XYZ[0] |= mSamples[1] * 256;
XYZ[0] |= mSamples[2];

XYZ[1] = ((signed char)mSamples[3]) * 256 * 256;
XYZ[1] |= mSamples[4] * 256;
XYZ[1] |= mSamples[5];

XYZ[2] = ((signed char)mSamples[6]) * 256 * 256;
XYZ[2] |= mSamples[7] * 256;
XYZ[2] |= mSamples[8];
```

7 HOW TO CONVERT RM3100 OUTPUT TO MICROTESLA?

Convert the output of RM3100 from count to microTesla, the equation is,

magnetic_field_in_uT = (RM3100_measurement_in_count / Gain)

Please pay attention to “Cycle Count” setting. The default is 200, if customer changes to different value, the “Gain” will change too. Example given in Q4 Table.

8 WHAT IS THE MAXIMUM SAMPLE OR DATA RATE?

The Cycle Count establish the maximum sample rate. For instance, if the cycle count is set to 200 as default, then the maximum 3-axis sample rate is ~430 Hz. Higher sample rates can be obtained by reducing the number of cycle counts, but this also diminishes gain and sensitivity. When sample rate is fixed, lower cycle count will reduce sensor oscillation time and in turn to reduce power consumption.

9 HOW TO CHANGE SAMPLE OR DATA RATE?

RM3100 TMRC register controls the time between measurements in Continuous Measurement Mode. Its default is 0x92, indicating an update rate of ~600 Hz. When the Cycle Count is 200, the maximum rate established by the Cycle Count will override the TMRC setting, and the actual update rate will be ~430 Hz.

10 HOW MANY MODES RM3100 SUPPORTS?

RM3100 supports continuous measurement mode and single measurement polling.

11 DOES RM3100 SUPPORT DATA READY SIGNAL?

Yes. RM3100 supports measurement data ready interrupt line (PIN 23) and also STATUS Register (0x34).

12 HOW TO READ RM3100 OUTPUT?

Assume SPI host or I2C host is well established to communicate to RM3100. Read RM3100 output from Measurement Results Registers 0x24 to 0x2C. There are total 9 bytes, and 3 bytes (24 bits) for each axis.

13 DOES RM3100 SUPPORT SELF TEST?

Yes. RM3100 supports Built-In Self-Test (BIST) with BIST register (0x33). For troubleshooting circuit connection on XYZ sensors, users should measure the resistance of XYZ sensors. The resistance value should be about 35 ohms. Big resistance value such as 1M ohms indicates broken sensor.

14 HOW TO EVALUATE RM3100?

PNI offers RM3100 Test Boards on-line, Eval Board and Break Board. They are the same PCB, and the Eval Board has extra pins for easy connection. Check RM3100 Testing Boards Manual for details.

15 ANY SAMPLE CODE OR DRIVER CODE ?

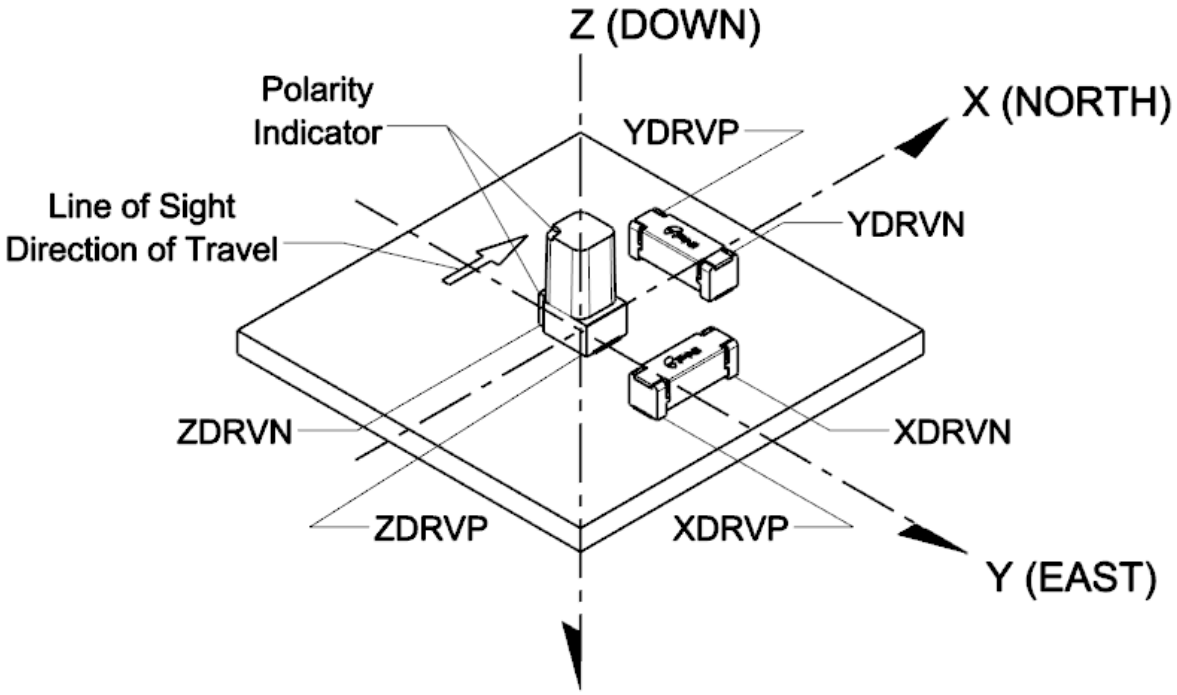
Sample code available on Arm Mbed OS platform.

I2C interface sample code can be found at the following link.
https://os.mbed.com/users/ddelsuc/code/RM3100BB_Sample_Code/

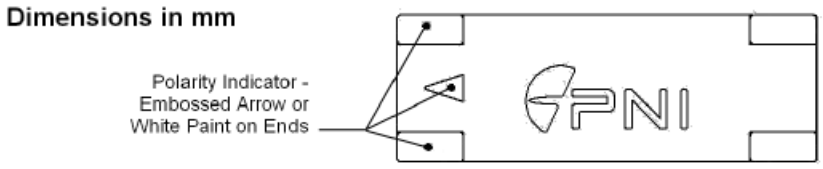
SPI interface sample code can be found at the following link.
https://os.mbed.com/users/BeiZhang/code/RM3100_SPI_SampleCode/

16 WHAT IS THE RECOMMENDED RM3100 LAYOUT AND AXIS POLARITIES ?

Following picture indicates how the three sensor coils in a RM3100 Geomagnetic Sensor should be oriented for a system referenced as north-east-down (NED) as XYZ axes. The Line of Sight arrow represents the direction of travel or pointing, and it's the X axis too. The Sen-XY-f is insensitive to the location of the polarity indicator, while the location of the polarity indicator is critical for the Sen-Z-f.



The polarity of X and Y axes are defined by the XDRVP and YDRVP pads. The embossed arrow or white paint on the XY coils in the following drawing can be ignored. However, PNI recommends to layout the XY coil's polarity indicator to be consistent on XDRVP and YDRVP for a clear layout design.



The polarity of Z axis is defined by both the ZDRVP and the Z coil polarity indicator. For the NED coordinate reference, the Z coil polarity indicator is on ZDRVN pad, therefore, the Z-axis positive polarity is "Down". If the Z coil is rotated 180° on the PCB, so that its polarity indictor to be on ZDRVP pad, the Z axis positive polarity will become "UP".

17 HOW TO TEST AND VERIFY POLARITY ON XYZ (NED) AXES ?

Geomagnetic field can be measured everywhere on earth. This example shows how to measure it to test and confirm the polarity of XYZ (NED) axes.

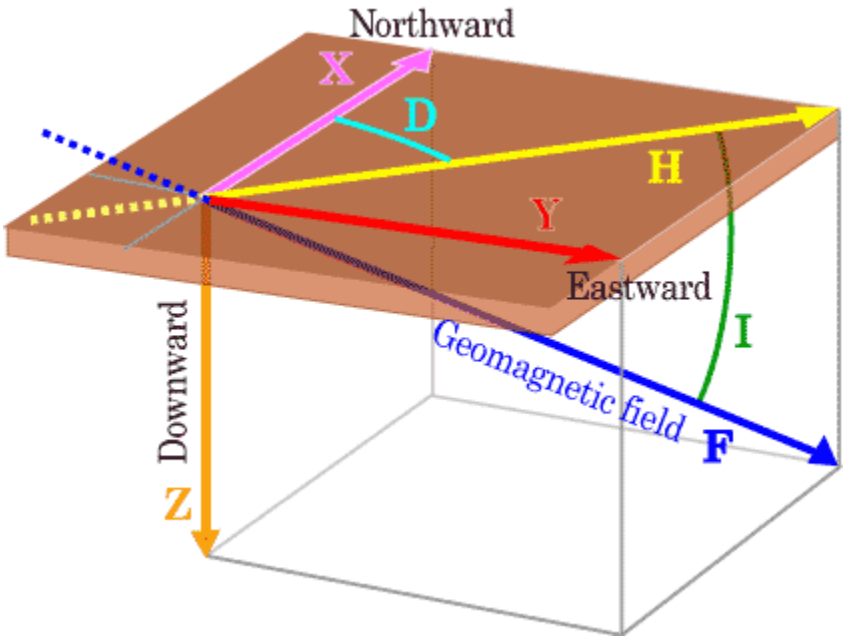
- Find a magnetic clean environment in office or outdoor. Move known magnetic distortion sources such as computers, smartphones, chairs, cabinets and any objects with ferrous material at least 6 ft (2m) away from RM3100.

- Find the longitude and latitude of your location.

For example, Lon: 122.755348, Lat: 38.420951, North Hemisphere, West

- Enter the Lon&Lat into the following calculator to get geomagnetic field strength and inclination of your location.
  https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml?model=wmm#igrfwmm
  Result: Geomagnetic field vector strength 48.35 μT, Inclination (dip angle) 61.43°

- Align the X axis toward the "Geomagnetic field F" vector like the following picture. Point X to north direction and tilt the PCB downward at the inclination angle of 61.43° if your location is at the northern hemisphere. When you are adjusting the alignment, observing the XYZ measurements, the measurement on X should be getting close to 48.35 μT. Y and Z measurement should be getting around 0. If X measurement is close to -48.35 μT, that means the polarity of X is reversed.



Where: I – Inclination, D – Declination, F – Geomagnetic field vector
(http://wdc.kugi.kyoto-u.ac.jp/element/eleexp.html)

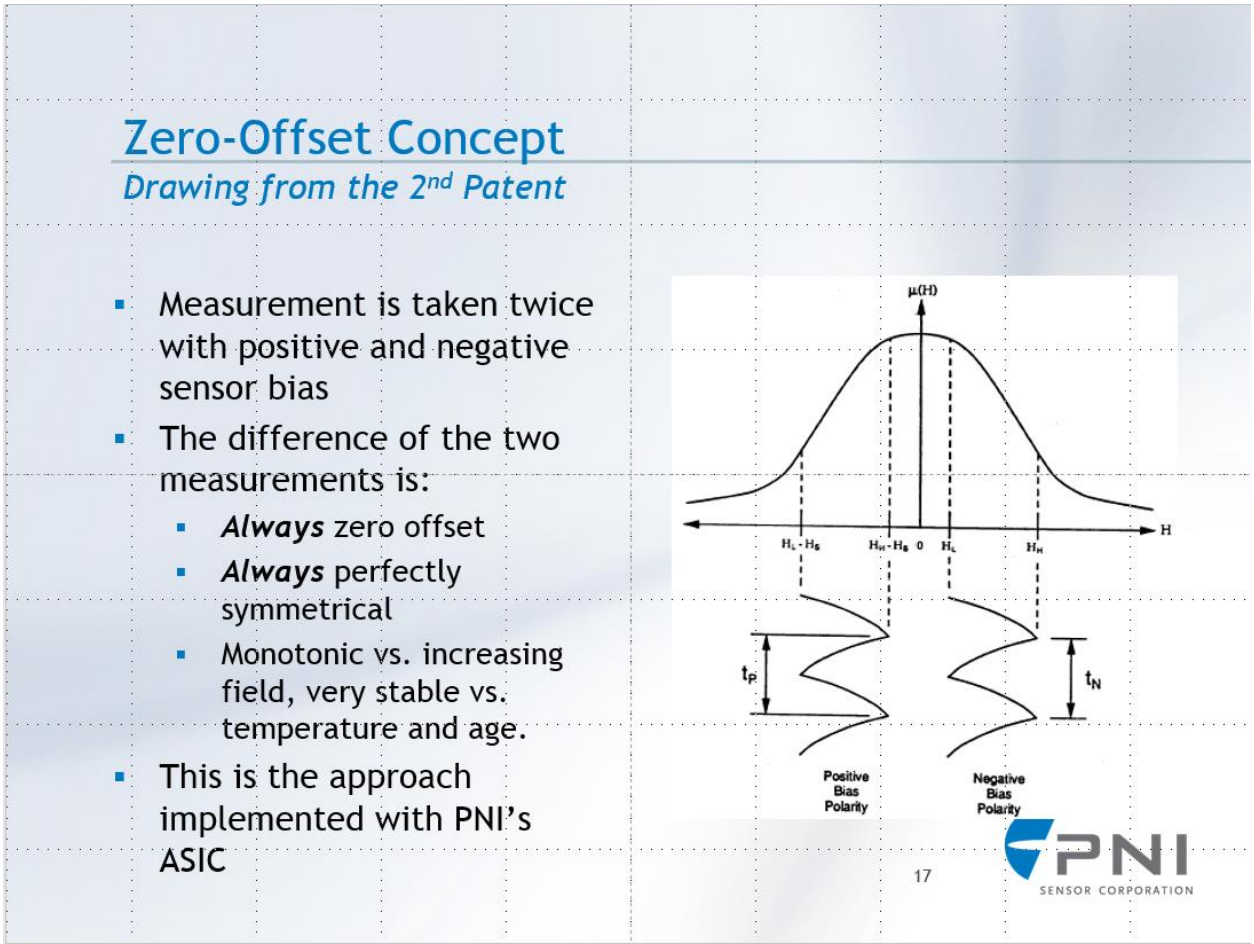
- Repeat the same procedure on Y. Align the Y axis toward the "Geomagnetic field F" vector, the measurement on Y should be around 48.35 μT. X and Z measurement should be around 0. If Y measurement is negative around 48.35 μT, that means the polarity of Y is reversed.
- Do the same test on Z to verify Z polarity.

18 WHAT IS RM3100 SENSOR ID?

REVID register 0x36 can be used to identity RM3100. Its readout value should be fixed as 0x21, 0x22 or higher, depending on when the sensor was made. The newer batch, the higher of the revision ID. RM3100 revision has not been changed since 2015.

19 DOES TEMPERATURE AFFECT RM3100 ?

RM3100 itself is not affected by temperature fluctuation. Because its LR oscillation circuit takes measurement twice with positive and negative sensor bias. The difference of two measurements as output cancels out temperature effect. See the following slide for your reference. Once RM3100 is integrated into a host device, the magnetic distortion sources introduced by the host device is affected by temperature changes, and that in turn will cause fluctuation in magnetic fields. RM3100 measures magnetic fields and surely its measurement will reflect the change. Therefore, the host device should still need to implement some level of temperature compensation.



Overview of PNI MI technology can be found here, https://www.pnicorp.com/download/pni-magneto-inductive-technology-overview/.

20 HOW TO START A MEASUREMENT?

RM3100 Testing Boards User Manual. Details in 5.7 Examples using the SPI Interface, and 5.8 Examples using the I2C Interface.

Summary to poll single measurement:

1. Set cycle count. (Optional, because you can use default 200 cycle count)

2. Host MCU initiates a Single measurement on all 3 axes to the POLL register 0x00.

3. Confirm New Data is Ready with Status Register 0x34 or check DRDY interrupt line.

4. Read Measurement Results registers 0x24 to 0x2C, 3 bytes for each axis.

21 WHAT IS THE RM3100 REGISTER MAP?

Following table is a full set of RM3100 registers, please read RM3100 user manual for details.

Name	Register # (Hex)	R/W	Default (Hex)	Payload Format	Description
POLL	00	RW	00	[7:0]	Polls for a Single Measurement
CMM	01	RW	00	[7:0]	Initiates Continuous Measurement Mode
CCX	04 – 05	RW	00C8	UInt16	Cycle Count Register – X Axis
CCY	06 – 07	RW	00C8	UInt16	Cycle Count Register – Y Axis
CCZ	08 – 09	RW	00C8	Uint16	Cycle Count Register – Z Axis
TMRC	0B	RW	96	[7:0]	Sets Continuous Measurement Mode Data Rate
ALLX	0C – 0E	RW	000000	Sint24	Alarm Lower Limit – X Axis
AULX	0F – 11	RW	000000	Sint24	Alarm Upper Limit – X Axis
ALLY	12 – 14	RW	000000	Sint24	Alarm Lower Limit – Y Axis
AULY	15 – 17	RW	000000	Sint24	Alarm Upper Limit – Y Axis
ALLZ	18 – 1A	RW	000000	Sint24	Alarm Lower Limit – Z Axis
AULZ	1B – 1D	RW	000000	Sint24	Alarm Upper Limit – Z Axis
ADLX	1E – 1F	RW	0000	SInt16	Alarm Hysteresis Value – X Axis
ADLY	20 – 21	RW	0000	SInt16	Alarm Hysteresis Value – Y Axis
ADLZ	22 – 23	RW	0000	SInt16	Alarm Hysteresis Value – Z Axis
MX	24 – 26	R	000000	Sint24	Measurement Results – X Axis
MY	27 – 29	R	000000	Sint24	Measurement Results – Y Axis
MZ	2A – 2C	R	000000	Sint24	Measurement Results – Z Axis
BIST	33	RW	00	[7:0]	Built-In Self-Test
STATUS	34	R	00	[7:0]	Status of DRDY
HSHAKE	35	RW	1B	[7:0]	Handshake Register
REVID	36	R	--	Unit8	MagI^{2}C Revision Identification

22 HOW TO READ/WRITE A REGISTER ON SPI?

RM3100 register addresses are 7 bits. To Write to a register on SPI interface, a '0' followed by the 7-bit register number should be sent, followed by the register value. ***To Read from a register, a '1' followed by the 7-bit register number should be sent, effectively adding 0x80 to the register number.***

Register Description	Default Value (Hex)	Write Address (Hex)	Read Address (Hex)
CCX - MSB	0x00	04	84
CCX - LSB	0xC8	05	85

23 HOW TO READ/WRITE A REGISTER ON I2C?

RM3100 register addresses are 7 bits. I2C interface uses the 7-bit address to Read or Write to a register.

Register Description	Default Value (Hex)	Read/Write Address (Hex)
CCX - MSB	0x00	04
CCX - LSB	0xC8	05